



RIVER HIGHER INSTITUTE OF TECHNOLOGY

ASSIGNMENTS 2025 SEMESTER 2

DIPLOMA IN INFORMATION TECHNOLOGY, LEVEL 6.

COMPUTER SYSTEMS ARCHITECTURE

Due Dates		
Assignment 1	29 August 2025	Semester 2
Assignment 2	12 September 2025	Semester 2

Dear students,

Welcome to **RIVER HIGHER INSTITUTE OF TECHNOLOGY** particularly to the business department. We hope your studies will be pleased with success. We advise that you get all the relevant information and student booklets through the learner management system provided by your lecturers. These documents will provide advice on how to approach your studies and will guide you through your study materials as well as providing useful administrative information.

Study material

Your study guide is essentially your "lecturer". However, in addition you are required to visit the library to consult prescribed books and recommended readings that are indicated in the study guide. You are also responsible to purchase any prescribed textbooks required for your course. Furthermore you are strongly advised to follow the syllabus guidelines.

Submission of Assignment

It is important to write and submit a clean, clear and well-argued assignment for marking. Make sure that you have answered the questions correctly. It is the responsibility of you as a student to make sure that every page of the completed assignment is numbered in the correct order before the pages are stapled together inside the assignment cover sheet. Assignment cover sheets are supplied on request from your lecturers. You should complete and on or before the given due date in the assignment letter if you wish to be admitted to the next examination.

Note that assignments received **after the due dates** will not be accepted and therefore will not be included in the calculation of continuous assessments (CA) marks. The CA weighted mark and the examination mark will determine your final mark for the course. Also please take note that assignments should never be submitted directly to individual lectures or any other staff members of the Institution, but directly to exam coordinators of the college within your centres.

When submitting your assignment electronically via email, make sure you use the correct email as indicated below. (**Only for distance students**). send to **exam@riverhigherinstitute.edu.na**.

Late Submission of Assignments or writing of tests

Late submission will not be marked. If there is a reason why you cannot meet the due date, please consider cancelling the module within the cancellation deadlines or contact your Head of centre for advice.

The following reasons may be considered for extension of due date:

- ❖ Personal illness or injury, or serious illness or death of an immediate relative, immediately preceding assignment due date, provided that a death certificate or a medical certificate specifying the nature and duration of the illness or injury is attached.
- ❖ Other circumstances that may affect your studies and that justify the late submission of an assignment or writing of a test

It is the responsibility of individual students to apply for an extension before the due dates of the assignments and submit supporting documentation.

Learner Management Portal

We recommend that you regularly access the learner management system on [https://, River Higher Institute of Technology E- learning](https://riverhighereducation.co.uk/learning).

Closing remarks

If you study conscientiously, your efforts will be rewarded. Should you need any assistance or clarification on the course content, you can contact the relevant staff members at the contact details as given below

Lecturer	Isaac Makosa	0817080991	makosaisaac@gmail.com
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Assignment 1: Understanding Number Systems and Data Representation

Title: *Binary Arithmetic and Data Representation in Computer Systems*

Objective:

To develop practical skills in performing binary arithmetic operations and representing different data types, including floating-point numbers and character encoding formats.

Instructions:

1. Define and explain the importance of binary number systems in computer systems.
2. Convert the following decimal numbers to:
 - Binary
 - Octal
 - Hexadecimal
 - (a) 156
 - (b) 45
 - (c) 1023
3. Perform the following binary arithmetic operations:
 - $1011 + 1101$
 - $11100 - 1101$
 - 101×11
4. Convert the following numbers into IEEE 754 floating point format:
 - 10.5
 - -5.75
5. Define the following terms with examples:
 - ASCII
 - Unicode
 - Signed Magnitude vs Two's Complement
6. Discuss data overflow and underflow with simple examples.
7. Explain the difference between 16-bit, 32-bit, and 64-bit word length. What are the implications for performance?

Deliverables: Typed document with explanations, workings, and final answers.

Total Marks: 50

Assignment 2: Architecture and Assembly Concepts

Title: *Basic Computer Architecture and Assembly Language Concepts*

Objective:

To analyse the structure of computer architecture, distinguish between processor types, and understand the basics of assembly language and CPU data handling.

Instructions:

1. Describe the **Von Neumann Architecture**. Draw and label its block diagram.
2. Explain the **Fetch-Execute cycle** with each step clearly shown.
3. Using a sample pseudocode, illustrate how data moves from **memory to CPU** and back.
4. Define the following CPU components and describe their roles:
 - ALU
 - Registers (General and Special Purpose)
 - Buses (Data, Address, Control)
 - Program Counter (PC)
 - Memory Address Register (MAR)
 - Memory Data Register (MDR)
5. Write a simple assembly-like program to:
 - Add two numbers
 - Store the result in memory
 - Show how this program would be interpreted by an assembler.
6. Compare CISC and RISC architectures:
 - Give 3 differences
 - Provide examples of processors that use each type
7. Explain the difference between **cache** and **virtual memory**.

Deliverables: Report including diagrams, sample codes, and explanations.

Total Marks: 50

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